Valuations

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1 Valuations subrings of a field

Definition 1.1. Let K be a field and L be an algebraically closed field. The *ordered set of homomorphisms from subrings of* K *into* L is the set of pairs (A, ϕ) where A is a subring of K and $\phi : A \to L$ is a homomorphism, equipped with the order where $(A, \phi) \leqslant (A', \phi')$ if and only if $A \subseteq A'$ and $\phi'|_A = \phi$.

Lemma 1.2. *Let* K *be a field and* L *be an algebraically closed field. Every nonempty chain in the ordered set of homomorphisms from subrings of* K *into* L *has an upper bound. Thus, given any subring* A *and a homomorphism* $\phi : A \to L$*, there exists a maximal element of the ordered set of homomorphisms from subrings of* K *into* L *which is greater than* (A, ϕ)*.*

Proof. Suppose I is a nonempty totally ordered set and we have a chain $(A_i, \phi_i)_{i \in I}$ indexed by I. Then $A = \bigcup_{i \in I} A_i$ is a subring of K containing A_i for all i, and defining $\phi : A \to L$ by declaring that $\phi(a) = \phi_i(a)$ for any $i \in I$ such that $a \in A_i$ yields a well defined homomorphism from A into L. The second assertion then follows immediately from Zorn's lemma. $\square$

Lemma 1.3. *Let K be a field and A a subring, and suppose that $x \in K$ is not integral over A. Then x^{-1} is not a unit in $A[x^{-1}]$. If $\mathfrak{m}$ is a maximal ideal of $A[x^{-1}]$ containing x^{-1}, then the composite $A \hookrightarrow A[x^{-1}] \to A[x^{-1}]/\mathfrak{m}$ is surjective, and $A \cap \mathfrak{m}$ is a maximal ideal in A.* [Bou72, VI.1, lemma 1]

Proof. If x^{-1} were a unit in $A[x]$, we have

$$x = a_0 + a_1 x^{-1} + \cdots + a_n x^{-n}$$

for some $a_0, \ldots, a_n \in A$. Multiplying through by x^n we see that x satisfies a monic polynomial of degree $n+1$ over A, contradicting our assumption that x not be integral over A. Thus x^{-1} cannot be a unit in $A[x^{-1}]$.

Now suppose that $\mathfrak{m}$ is a maximal ideal of $A[x^{-1}]$ containing x^{-1}. Then $A[x^{-1}]/\mathfrak{m}$ is a field, and the composite $A \hookrightarrow A[x^{-1}] \to A[x^{-1}]/\mathfrak{m}$ clearly has kernel $\mathfrak{m} \cap A$. Moreover, this composite is surjective since $x^{-1} \in \mathfrak{m}$, so $\mathfrak{m} \cap A$ must be a maximal ideal in A. □

Theorem 1.4. *Let K be a field and A a subring of K. Then the following are equivalent.* [Bou72, VI.1, theorem 1]

(a) *A is maximal in the set of local subrings of K ordered by domination.*

(b) *There exists an algebraically closed field L and a homomorphism $A \to L$ which is maximal in the ordered set of homomorphisms from subrings of K into L.*

(c) *If $x \in K \setminus A$ then $x^{-1} \in A$.*

(d) $\mathrm{Frac}(A) = K$, *A is local, and every finitely generated ideal of A is principal.*

(e) $\mathrm{Frac}(A) = K$ *and the set of principal ideals of A is totally ordered by inclusion.*

(f) $\mathrm{Frac}(A) = K$ *and the set of ideals of A is totally ordered by inclusion.*

Proof. (a) $\implies$ (b). Note that A is a local ring with maximal ideal $\mathfrak{m}$. Let L an algebraic closure of the residue field $A/\mathfrak{m}$, and $A \to L$ the natural map. Notice that its kernel is precisely $\mathfrak{m}$. Suppose B is a subring of K containing A and $\phi : B \to L$ a homomorphism extending ϕ. Then $\ker(\phi)$ is a prime ideal of B and $A \cap \ker(\phi) = \mathfrak{m}$. Then the local ring $B_{\ker(\phi)}$ dominates A, so we must have $A = B_{\ker(\phi)}$. Thus ϕ coincides with $A \to L$, proving that $A \to L$ is maximal in the set of homomorphisms from subrings of A into L ordered by extension.

(b) $\implies$ (c). Let L be a field and $\phi : A \to L$ maximal in the set of homomorphisms from subrings of A into L. Then $\ker(\phi)$ is a prime ideal of A and ϕ can be extended to the local ring $A_{\ker(\phi)}$ which is a subring of K containing A. By maximality, we must have $A = A_{\ker(\phi)}$, so A is a local ring and $\ker(\phi)$ is its maximal ideal.

Suppose we have $x \in K$ is integral over A, then there exists a monic polynomial $f = T^n + a_1 T^{n-1} + \cdots + a_n \in A[T]$ for some formal indeterminate T such that $T \mapsto x$ defines an isomorphism of A-algebras $A[T]/(f) = A[x]$. Since L is algebraically closed, $\phi(f)$

has a root $\alpha \in L$, so $x \mapsto \alpha$ defines an A-algebra homomorphism $A[x] \to L$. By maximality, we must have $A[x] = A$. In other words, A is integrally closed in K.

Now suppose that $x \in K \setminus A$. Then, as we have just seen, x is not integral over A. Lemma 1.3 tells us thatx^{-1} is not a unit in $A[x^{-1}]$, so there exists a maximal ideal $\mathfrak{m}$ of $A[x^{-1}]$ containing x^{-1}. Then $\mathfrak{m} \cap A$ is a maximal ideal of A, but A is local with maximal ideal $\ker(\phi)$, so $\ker(\phi) = \mathfrak{m} \cap A$ is the kernel of the surjection $A \to A[x^{-1}]/\mathfrak{m}$. Thus ϕ naturally extends to a homomorphism $A[x^{-1}]/\mathfrak{m} \to L$. Precomposing with the quotient map $A[x^{-1}] \to A[x^{-1}]/\mathfrak{m}$ yields an extension of ϕ to $A[x^{-1}]$. By maximality, we conclude that $A[x^{-1}] = A$. In other words, we have $x^{-1} \in A$.

(c) $\implies$ (d). It is clear that $\mathrm{Frac}(A) = K$. Let us show that A is local. We know that A has at least one maximal ideal $\mathfrak{m}$. Suppose it had a second maximal ideal $\mathfrak{m}'$. Then $\mathfrak{m}' \not\subseteq \mathfrak{m}$ and $\mathfrak{m} \not\subseteq \mathfrak{m}'$, so there exists $x \in \mathfrak{m} \setminus \mathfrak{m}'$ and $x' \in \mathfrak{m}' \setminus \mathfrak{m}$. Now consider $x/x' \in K$. If $x/x' \in A$, then $x = x'(x/x') \in \mathfrak{m}'$, contradicting our choice of x, so $x/x' \notin A$. But for the same reasons, we also have $x'/x \notin A$. This is a contradiction. Thus A must have just one maximal ideal.

Finally, to see that every finitely generated ideal is principal, it is sufficient to show that any ideal $I = (x, y)$ that is generated by two elements $x, y \in A$ must be principal. If $y = 0$, then $I = (x)$ and we are done, so suppose $y \neq 0$ and consider $x/y \in K$. If $x/y \in A$, then $x = (x/y)y$ so $I = (y)$ and we are done. Otherwise we have $y/x \in A$, and then $y = (y/x)x$ so $I = (x)$ and again we are done.

(d) $\implies$ (e). Suppose we have two elements $x, y \in A$. We want to show that either $(x) \subseteq (y)$ or $(y) \subseteq (x)$. Consider the ideal $I = (x, y)$. Since I is principal, there exist $a, b \in A$ such that $I = (ax + by)$. Moreover, there exists $u \in A$ such that $x = u(ax + by)$.

If u is a unit, then $(x) = (ax + by) = I \supseteq (y)$ and we are done. If u is not a unit, then u is in the maximal ideal $\mathfrak{m}$ of A. Then $x(1 - ua) = uby$. Since $u \in \mathfrak{m}$, then $ua \in \mathfrak{m}$, so $1 - ua$ is a unit in A, so $x = (1 - ua)^{-1}uby \in (y)$, proving that $(x) \subseteq (y)$.

(e) $\implies$ (f). Suppose I and J are ideals of A and $J \not\subseteq I$. Then there exists $y \in J \setminus I$, which means that for any $x \in I$, we have $y \notin (x)$. In other words, $(y) \not\subseteq (x)$. Thus $(x) \subseteq (y)$. Since $x \in I$ is arbitrary, we see that $I \subseteq (y) \subseteq J$.

(f) $\implies$ (a). Since the ideals of A are totally ordered, it must be a local ring. Suppose B is a local subring of K dominating A. Then for any $z \in B \subseteq K = \mathrm{Frac}(A)$, we can write $z = x/y$ with $x, y \in A$. We want to show that $z \in A$.

Suppose for a contradiction that $z = x/y \notin A$. This means that $(x) \subsetneq (y)$, so since the ideals of A are totally ordered by inclusion, we must have $(y) \subseteq (x)$. In other words, $z^{-1} = y/x \in A$. Since $z \notin A$ by assumption, we see that z^{-1} is a non-unit in A, which means that z^{-1} is in the maximal ideal of A. On the other hand, we know that $z \in B$ and $z^{-1} \in A \subseteq B$, so z is a unit in B. This means that z^{-1} cannot be in the maximal ideal of B, which contradicts the assumption that B dominates A. $\square$

Definition 1.5. Let K be a field. A *valuation subring* of K is a subring satisfying one of the equivalent conditions of theorem 1.4.

Lemma 1.6. *Let A be a valuation subring of a field K. Then A is integrally closed in K.*

Proof. This is shown during the proof of theorem 1.4, (b) $\implies$ (c) above. $\square$

Lemma 1.7. *Let A be a local domain and $K = \mathrm{Frac}(A)$. Then there exist a valuation subring of K dominating A.* [Bou72, VI.1, corollary to theorem 2]

Proof. Let L be an algebraic closure of the residue field of A and $\phi : A \to L$ the canonical map. Then $\ker(\phi)$ is the maximal ideal of A. The ordered set of homomorphisms from subrings of K into L is nonempty since it contains (A, ϕ), so it contains a maximal element (A', ϕ') extending (A, ϕ) by lemma 1.2. Then A' is a valuation subring of K by condition (b) of theorem 1.4. Note that $\ker(\phi')$ is a prime ideal of A' and ϕ' can be extended naturally to $A'_{\ker(\phi')}$, so maximality guarantees that $A' = A'_{\ker(\phi')}$. In other words, $\ker(\phi')$ is the maximal ideal of A'. Now clearly $A \cap \ker(\phi') = \ker(\phi)$ since ϕ' extends ϕ, so we see that A' dominates A. $\square$

Lemma 1.8. *Let A be a domain and $K = \mathrm{Frac}(A)$. The integral closure of A in K is the intersection of all valuation subrings of K containing A.* [Bou72, VI.1, theorem 3]

Proof. Let A' denote the integral closure of A in K and let B be a valuation subring of K containing A. If $x \in A'$, then x is integral over A and therefore also over B. Thus the inclusion $B \hookrightarrow B[x]$ is a finite homomorphism, so by going up [Stacks, 00GQ] we find that there exists a prime ideal $\mathfrak{p} \subseteq B[x]$ such that $\mathfrak{p} \cap B$ is the maximal ideal $\mathfrak{m}_B$ of B. Then $B[x]_{\mathfrak{p}}$ is a local ring dominating B, so condition (a) of theorem 1.4 guarantees that $B = B[x]_{\mathfrak{p}}$. In particular, we see that $x \in B$, so $A' \subseteq B$.

Conversely, suppose we have $x \in K \setminus A'$. In other words, x is not integral over A, so lemma 1.3 tells us that x^{-1} is not a unit in $A[x^{-1}]$, so there exists a maximal ideal $\mathfrak{m}$ of $A[x^{-1}]$ containing x^{-1} and that $\mathfrak{m} \cap A$ is a maximal ideal of A. Then there exists a valuation subring B of K dominating $A[x^{-1}]_{\mathfrak{m}}$ by lemma 1.7. Since x^{-1} is not a unit in $A[x^{-1}]_{\mathfrak{m}}$, it is not a unit in the dominating ring B either. Thus $x \notin B$. $\square$

Lemma 1.9. *Let K be a field and A a valuation subring of K. If B is a subring of K containing A, then B is a valuation subring of K, and the maximal ideal $\mathfrak{m}_B$ of B is a prime ideal of A. The map $B \mapsto \mathfrak{m}_B$ is an inclusion reversing bijection between the set of subrings of K containing A onto $\mathrm{Spec}(A)$, with inclusion reversing inverse given by $\mathfrak{p} \mapsto A_{\mathfrak{p}}$.* [Bou72, VI.4, proposition 1]

Proof. We will use condition (c) of theorem 1.4 as a characterization of valuation subrings of K repeatedly. Suppose B is a subring of K containing A. If $x \in K \setminus B$, then $x \in K \setminus A$, so $x^{-1} \in A \subseteq B$, so B is a valuation ring. Moreover, if $x \in \mathfrak{m}_B$, then $x^{-1} \in K \setminus B \subseteq K \setminus A$, so $x \in A$. In other words, $\mathfrak{m}_B \subseteq A$. Then $\mathfrak{m}_B = \mathfrak{m}_B \cap A$, so $\mathfrak{m}_B$ must be a prime ideal of A.

Since every element of $A \setminus \mathfrak{m}_B$ is a unit in B, we have $A_{\mathfrak{m}_B} \subseteq B$. If $x \in B \setminus A_{\mathfrak{m}_B}$, then $x \notin A$ and $x^{-1} \notin \mathfrak{m}_B$. But, since A is a valuation ring, we do have $x^{-1} \in A$. In other words, $x^{-1} \in A \setminus \mathfrak{m}_B$, which means that $x \in \mathfrak{m}_B$. Thus $B = A_{\mathfrak{m}_B}$.

If $\mathfrak{p}$ is a prime ideal of A, then $A_{\mathfrak{p}}$ is a subring of K containing A, so it is a valuation subring of K, and its maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ is contained in A and must equal $\mathfrak{p}$. This shows that $\mathfrak{p} \mapsto A_{\mathfrak{p}}$ is bijective and that its inverse is $B \mapsto \mathfrak{m}_B$.

The map $\mathfrak{p} \mapsto A_{\mathfrak{p}}$ is clearly inclusion reversing. We know from condition (f) of theorem 1.4 that $\mathrm{Spec}(A)$ is totally ordered by inclusion, so it follows from ordered-groups, lemma 1.5 that its inverse must also be an inclusion reversing. □

Lemma 1.10. *Let K be a field, A a valuation subring of K, and K' a field extension of K. Then there exists a valuation subring A' of K' such that $A' \cap K = A$. In fact, if $x_1, \ldots, x_n$ are algebraically independent elements of K' over K, then A' can be chosen so that $A[x_1, \ldots, x_n] \subseteq A'$.*

Proof. Let L be an algebraic closure of the residue field of A and $\phi : A \to L$ the canonical map. Since $x_1, \ldots, x_n$ are algebraically independent over K, the ring $A[x_1, \ldots, x_n]$ is a polynomial algebra over A. So, by choosing an arbitrary n-tuple of elements of L, we can extend ϕ to a homomorphism $\phi : A[x_1, \ldots, x_n] \to L$.

Now the pair $(A[x_1, \ldots, x_n], \phi)$ defines an element of the ordered set of homomorphisms from subrings of K' into L. Applying lemma 1.2, we can enlarge $A[x_1, \ldots, x_n]$ to a maximal subring A' of K' which contains $A[x_1, \ldots, x_n]$ such that ϕ extends to A'. By condition (b) of theorem 1.4, we know that A' is a valuation subring of K'.

We know that $A \subseteq A' \cap K$ immediately. Moreover, observe that $A' \cap K$ is a subring of K and that $\phi : A' \to L$ restricts to a homomorphism $A' \cap K \to L$ which extends the original map $A \to L$. In other words, the pair $(A' \cap K, \phi)$ is an element of the ordered set of homomorphisms from subrings of K into L which is larger than (A, ϕ). Since A is a valuation subring of K, we use condition (b) of theorem 1.4 once again to conclude that $A = A' \cap K$. □

2 Valuations on a ring

Definition 2.1. Suppose Γ is a totally ordered abelian group (cf. ordered-groups, definition 2.4). We let ∞ denote an element disjoint from Γ and $\Gamma \cup \{\infty\}$ denote the totally ordered set in which $\infty \geqslant x$ for all $x \in \Gamma$. We also extend the group operation $+$ on Γ to an operation on $\Gamma \cup \{\infty\}$ by declaring $x + \infty = \infty + x = \infty$ for all $x \in \Gamma \cup \{\infty\}$. This makes $\Gamma \cup \{\infty\}$ a totally ordered monoid. Also, for any $x \in \Gamma$, we define $\infty - x = \infty$.

Definition 2.2. Let A be a ring.

- A *valuation* on A is a function $v : A \to \Gamma \cup \{\infty\}$ for some totally ordered abelian group Γ which satisfies the following conditions for all $a, b \in A$.

  (1) $v(ab) = v(a) + v(b)$.

  (2) $v(a + b) \geqslant \min\{v(a), v(b)\}$.

  (3) $v(1) = 0$ and $v(0) = \infty$.

- If v is a valuation on A, then we associate to v the following subsets of A.

$$\begin{aligned} A^{v \geqslant 0} &= \{x \in A : v(x) \geqslant 0\} \\ A^{v \gneqq 0} &= \{x \in A : v(x) \gneqq 0\} \\ \operatorname{supp}(v) &= \{x \in A : v(x) = \infty\}. \end{aligned}$$

  We will see in lemma 2.4 that $\operatorname{supp}(v)$, called the *support of* v, is a prime ideal of A, and we define $\kappa(v) = \operatorname{Frac}(A/\operatorname{supp}(v))$.

- If $v : A \to \Gamma \cup \{\infty\}$ is a valuation on A, the subgroup of Γ generated by $v(A \setminus \operatorname{supp}(v))$ is called the *value group* of v and is denoted Γ_v.

Lemma 2.3. *Let A be a ring and v a valuation on A. Then $A^{v \geqslant 0}$ is a subring of A, and $A^{v \gneqq 0}$ is a proper ideal of $A^{v \geqslant 0}$.*

Proof. This is immediate from the axioms for a valuation in definition 2.2. □

Lemma 2.4. *Let A be a ring and v a valuation on A. Then $\operatorname{supp}(v)$ is a prime ideal in A.*

Proof. We know that $0 \in \operatorname{supp}(v)$, and if $a, b \in \operatorname{supp}(v)$, then

$$\infty \geqslant v(a+b) \geqslant \min\{v(a), v(b)\} = \infty$$

so $a + b \in \operatorname{supp}(v)$, proving that $\operatorname{supp}(v)$ is an additive subgroup of A. If $a \in \operatorname{supp}(v)$ and $x \in A$, then $v(ax) = v(a) + v(x) = \infty$ so $ax \in \operatorname{supp}(v)$, proving that $\operatorname{supp}(v)$ is an ideal of A. Note that $1 \notin \operatorname{supp}(v)$, so $\operatorname{supp}(v)$ is a proper ideal. Finally, suppose we have $a, b \in A$ such that $a \notin \operatorname{supp}(v)$ and $ab \in \operatorname{supp}(v)$. Then

$$\infty = v(ab) = v(a) + v(b)$$

and $v(a) \neq \infty$ so we must have $v(b) = \infty$. □

Lemma 2.5. *Let A be a ring and v a valuation on A. If $a \in A^\times$, then $v(a) \neq \infty$ and $v(a) = -v(a^{-1})$.*

Proof. Since $\operatorname{supp}(v)$ is a prime ideal by lemma 2.4, in particular it is a proper ideal so we must have $v(a) \neq \infty$. Moreover, $0 = v(1) = v(aa^{-1}) = v(a) + v(a^{-1})$. □

Lemma 2.6. *Let A be a ring and v a valuation on A. If $\zeta \in A$ is a root of unity, then $v(\zeta) = 0$.*

Proof. If $\zeta^n = 1$ for some $n \geqslant 1$, then $0 = v(1) = v(\zeta^n) = nv(\zeta)$. But totally ordered abelian groups are torsion free (cf. ordered-groups, lemma 2.5), so this forces $v(\zeta) = 0$. □

Lemma 2.7. *Let A be a ring and v a valuation on A. Then $v(a) = v(-a)$ for all $a \in A$.*

Proof. Since $(-1)^2 = 1$, it follows from lemma 2.6 that $\nu(-1) = 0$. Thus

$$\nu(-a) = \nu((-1)a) = \nu(-1) + \nu(a) = \nu(a). \qquad \square$$

Lemma 2.8. *Let A be a ring and ν a valuation on A. If we have $a, b \in A$ and $\nu(a) \neq \nu(b)$, then*

$$\nu(a + b) = \min\{\nu(a), \nu(b)\}.$$

Proof. We can assume without loss of generality that $\nu(a) \gneqq \nu(b)$. Then $\min\{\nu(a), \nu(b)\} = \nu(b)$. We know that $\nu(a + b) \geqslant \nu(b)$, and we want to show that $\nu(a + b) = \nu(b)$. Observe that

$$\nu(b) = \nu((a + b) - a) \geqslant \min\{\nu(a + b), \nu(a)\},$$

where we have used lemma 2.7. If $\min\{\nu(a + b), \nu(a)\} = \nu(a)$, then this tells us that $\nu(b) \geqslant \nu(a)$, contradicting our initial assumption. Thus it must be the case that $\nu(a) \gneqq \nu(a + b)$, and we have

$$\nu(b) \geqslant \min\{\nu(a + b), \nu(a)\} = \nu(a + b) \geqslant \nu(b),$$

forcing equality throughout. $\square$

Lemma 2.9. *Let $\phi : A \to B$ be a homomorphism of rings and ν a valuation on B. Then $\nu \circ \phi$ is a valuation on A.*

Proof. Omitted. $\square$

Lemma 2.10. *Let A be a field and $\nu : A \to \Gamma \cup \{\infty\}$ a valuation on A. Then there exists a unique valuation on $\kappa(\nu) = \mathrm{Frac}(A/\operatorname{supp}(\nu))$ which makes the following diagram commute.*

$$\begin{tikzcd} A \arrow[r] \arrow[dr, bend right, "\nu"'] & \kappa(\nu) \arrow[d, "\nu"] \\ & \Gamma \cup \{\infty\} \end{tikzcd}$$

Moreover, the value group of $\nu : A \to \Gamma \cup \{\infty\}$ coincides with $\nu(\kappa(\nu) \setminus \{0\})$.

Proof. First note that there exists a unique valuation $\bar{\nu}$ on $A/\operatorname{supp}(\nu)$ which makes the following diagram commute.

$$\begin{tikzcd} A \arrow[r, "\pi"] \arrow[dr, bend right, "\nu"'] & A/\operatorname{supp}(\nu) \arrow[d, "\bar{\nu}"] \\ & \Gamma \end{tikzcd}$$

Uniqueness is clear, since the quotient map $\pi : A \to A/\operatorname{supp}(\nu)$ is surjective. We need to prove existence. Suppose we have $a = b + \epsilon$ where $\epsilon \in \operatorname{supp}(\nu)$. We claim that $\nu(a) = \nu(b)$.

If $b \in \operatorname{supp}(\nu)$, then $a \in \operatorname{supp}(\nu)$ by lemma 2.4 and we are done. Otherwise, if $b \notin \operatorname{supp}(\nu)$, then $\nu(b) \lneq \nu(\epsilon)$, so by lemma 2.8 we see that $\nu(a) = \nu(b)$. Thus there exists a well-defined function $\bar{\nu}$ which makes the above diagram commute. Now it follows from the fact that ν is a valuation that $\bar{\nu}$ must also be a valuation.

Thus, by replacing A with $A/\operatorname{supp}(\nu)$, we can assume that A is a domain and $\operatorname{supp}(\nu) = 0$. Let $K = \operatorname{Frac}(A)$. We now want to show that ν extends uniquely to a valuation $\bar{\nu}$ on K making the following diagram commute.

$$\begin{tikzcd} A \arrow[r, hook] \arrow[dr, bend right, "\nu"'] & K \arrow[d, "\bar{\nu}"] \\ & \Gamma \end{tikzcd}$$

For uniqueness, note simply that every element of K can be written as a/b where $a, b \in A$ and $b \neq 0$, so if we had a valuation $\bar{\nu}$ on K extending ν, we would have to have

$$\bar{\nu}(a/b) = \nu(a) - \nu(b) \tag{2.11}$$

using lemma 2.5. To prove existence, we need to show that a function $\bar{\nu}$ defined as in equation (2.11) is a well-defined valuation on K.

First, let us show that $\bar{\nu}$ is a well-defined function. Suppose $a/b = a'/b'$ with $a, a', b, b' \in A$ and $b, b' \neq 0$. Then $ab' = a'b$ in A, so

$$\nu(a) + \nu(b') = \nu(a') + \nu(b).$$

Rearranging shows that $\nu(a') - \nu(b') = \nu(a) - \nu(b)$, proving that $\bar{\nu}(a/b) = \nu(a) - \nu(b)$ is a well-defined function on K.

It is clear that $\bar{\nu}$ extends ν. In particular, we have $\bar{\nu}(0) = \infty$. Next, let us show that $\bar{\nu}$ behaves well with respect to multiplication. Suppose we have a/b and a'/b'. Then

$$\bar{\nu}((a/b)(a'/b')) = \bar{\nu}(aa'/bb') = \nu(a) + \nu(a') - \nu(b) - \nu(b') = \bar{\nu}(a/b) + \bar{\nu}(a'/b').$$

Finally, we need to show that $\bar{\nu}$ behaves well with respect to addition.

$$\begin{aligned} \bar{\nu}((a/b) + (a'/b')) &= \nu(ab' + a'b) - \nu(bb') \\ &\geqslant \min\{\nu(ab'), \nu(a'b)\} - \nu(bb') \\ &= \min\{\nu(ab') - \nu(bb'), \nu(a'b) - \nu(bb')\} \\ &= \min\{\bar{\nu}(a/b), \bar{\nu}(a'/b')\}. \end{aligned}$$

Thus the function $\bar{\nu}$ is in fact a valuation.

□

3 Quotient and restricted valuation

Definition 3.1. Let A be a ring, $\nu : A \to \Gamma \cup \{\infty\}$ a valuation on A, and Δ a convex subgroup of Γ (cf. ordered-groups, definition 4.1). Then the quotient group Γ/Δ is naturally a totally ordered abelian group (cf. ordered-groups, lemma 6.1), and we define $\nu_{/\Delta}$ to be the composite

$$A \xrightarrow{\nu} \Gamma \cup \{\infty\} \longrightarrow \Gamma/\Delta \cup \{\infty\}.$$

Evidently, $\nu_{/\Delta}$ is a valuation on A.

Lemma 3.2. *Let A be a ring, $\nu : A \to \Gamma \cup \{\infty\}$ a valuation on A and Δ a convex subgroup of Γ. Then* $\operatorname{supp}(\nu) = \operatorname{supp}(\nu_{/\Delta})$.

Proof. This is clear from construction. □

4 Valuations on a field

Lemma 4.1. *Let K be a field and ν a valuation on K. Then $K^{\nu \geqslant 0}$ is a valuation subring of K and $K^{\nu \gneqq 0}$ is its maximal ideal. In particular, the group of units of $K^{\nu \geqslant 0}$ is precisely the set*

$$\{x \in K : \nu(x) = 0\}.$$

Proof. We know already from lemma 2.3 that $A = K^{\nu \geqslant 0}$ is a subrbing of K and that $\mathfrak{m} = K^{\nu \gneqq 0}$ is an ideal of A. If $x \in K^\times$, it follows from lemma 2.5 that either $\nu(x) \geqslant 0$ or $\nu(x^{-1}) \geqslant 0$. In other words, A satisfies condition (c) of theorem 1.4, so it is a valuation subring of K. Now note that x is a unit in A if and only if both x and x^{-1} are in A, if and only if $\nu(x) = 0$. Thus $\mathfrak{m}$ is precisely the set of non-units of A, so it is the maximal ideal of A. □

Lemma 4.2. *Let K be a field and $\nu : K \to \Gamma \cup \{\infty\}$ a valuation on K. Then $\Gamma_\nu = \nu(K^\times)$.*

Proof. Since K is a field and $\operatorname{supp}(\nu)$ is a prime ideal of K by lemma 2.4, it must be that $\operatorname{supp}(\nu) = 0$. Thus Γ_ν is the subgroup of Γ generated by $\nu(K^\times)$. But note that $\nu : K^\times \to \Gamma$ is a group homomorphism, so $\nu(K^\times)$ is already a subgroup of Γ. □

Lemma 4.3. *Let K be a field and A a valuation subring of K. Then there exists a totally ordered abelian group Γ and a surjective valuation $\nu : K \to \Gamma \cup \{\infty\}$ such that $K^{\nu \geqslant 0} = A$.*

Proof. Define $\Gamma = K^\times/A^\times$ with group law written additively. For any $x \in K^\times$, we let $\nu(x)$ denote its class in Γ. Then $\nu : K^\times \to \Gamma$ is a group homomorphism (it transforms multiplication into addition, and $\nu(a) = 0$ for any $a \in A^\times$).

Observe that $A \setminus \{0\}$ is a multiplicative submonoid of $K^\times$, so its image $S = \nu(A \setminus \{0\})$ in Γ is a submonoid of Γ. For any $x \in K^\times$, observe that $\nu(x) \in S$ if and only if $x \in A \setminus \{0\}$.

Indeed, the "if" direction is clear, and conversely, if $\nu(x) \in S$, then there exists some $u \in A^\times$ such that $ux \in A \setminus \{0\}$. But then $x = u^{-1}ux \in A \setminus \{0\}$ also. Let us show that S satisfies the conditions of ordered-groups, lemma 2.2. Suppose we have some $x \in A \setminus \{0\}$ such that $-\nu(x) \in S$. Then $-\nu(x) = \nu(x^{-1})$, so $x^{-1} \in A \setminus \{0\}$. In other words, $x \in A^\times$, so $\nu(x) = 0$. This proves that S satisfies ordered-groups, condition (1). Now ordered-groups, condition (2) follows immediately from the definition of valuation subrings. Thus S is the set of positive elements for a unique translation invariant total order on Γ.

We now extend ν to a function $K \to \Gamma \cup \{\infty\}$ by setting $\nu(0) = \infty$. We need to show that this is a valuation. We already know that ν is a group homomorphism when restricted to $K^\times$. Since $\nu(x \cdot 0) = \nu(0) = \infty$ for all $x \in K$, it follows that ν behaves well with respect to multiplication.

To see that it behaves well with respect to addition, suppose we have $a, b \in K$ and let us assume without loss of generality that $\nu(a) \geqslant \nu(b)$. We want to show that $\nu(a+b) \geqslant \nu(b)$. If either $a = 0$ or $b = 0$, then we have $\nu(a+b) = \nu(b)$ and we are done. If $a + b = 0$, we are again done immediately.

So we can assume that a, b and $a + b$ are all nonzero. Since $\nu(a) \geqslant \nu(b)$, we see that $\nu(a/b) \in A \setminus \{0\}$. Moreover, $(a+b)/b$ is nonzero since $a + b$ is nonzero, and

$$\frac{a+b}{b} = \frac{a}{b} + 1$$

is an element of A, since a/b is. This means that

$$\nu(a+b) - \nu(b) = \nu((a+b)/b) \geqslant 0,$$

completing the proof that ν is a valuation. It is clear from construction that $K^{\nu \geqslant 0} = A$. □

Lemma 4.4. *Let K be a field and $\nu : A \to \Gamma \cup \{\infty\}$ a valuation on K. Let $A = K^{\nu \geqslant 0}$ and let B be a subring of K containing A. If $x \in K^\times$ is such that $\nu(x) \in \nu(B^\times)$, then in fact $x \in B^\times$.*

Proof. Since $\nu(x) \in \nu(B^\times)$, there exists $y \in B^\times$ such that $\nu(x) = \nu(y)$. Then $\nu(x/y) = \nu(x) - \nu(y) = 0$, so $x/y \in A^\times$ by lemma 4.1. Thus $x/y \in B^\times$ also, so $x = y(x/y) \in B^\times$. □

Lemma 4.5. *Let K be a field and $\nu : K \to \Gamma \cup \{\infty\}$ a surjective valuation on K. Let $A = K^{\nu \geqslant 0}$. For a convex subgroup $\Delta \subseteq \Gamma$, we have the quotient valuation $\nu_{/\Delta}$ (cf. definition 3.1), and* [Bou72, VI.4, proposition 4]

$$B_\Delta = \{x \in K \ : \ \nu_{/\Delta}(x) \geqslant 0\}$$

is a subring of K containing A. Moreover, the map $\Delta \mapsto B_\Delta$ is an order isomorphism from $\mathrm{Cnv}(\Gamma)$ onto the set of subrings of K containing A ordered by inclusion. Its inverse maps a subring B of K containing A to the convex subgroup $\Delta_B = \nu(B^\times)$.

Proof. Since $\nu_{/\Delta}$ is a valuation on K, we know that B_Δ is a subring of K by lemma 2.3, and it contains A because the quotient map $\Gamma \to \Gamma/\Delta$ is order preserving (cf. ordered-groups,

lemma 6.1). If $\Delta' \subseteq \Delta$ are two convex subgroups of Γ, then Δ/Δ' is a convex subgroup of Γ/Δ' (cf. ordered-groups, lemma 6.3), so the quotient map

$$\Gamma/\Delta' \longrightarrow \Gamma/\Delta = (\Gamma/\Delta')/(\Delta/\Delta')$$

is also order preserving. Thus for any $x \in K$ we see that $v_{/\Delta'}(x) \geqslant 0$ implies $v_{/\Delta}(x) \geqslant 0$. In other words, we have $B_{\Delta'} \subseteq B_\Delta$, proving that $\Delta \mapsto B_\Delta$ is order preserving.

To see that it is injective, let us show that Δ coincides with $v(B_\Delta^\times)$ and can therefore be recovered from the subring B_Δ. Since v is surjective, every element of Γ can be written as $v(x)$ for some $x \in K^\times$. Then $v(x) \in v(B_\Delta^\times)$ if and only if $x \in B_\Delta^\times$ (cf. lemma 4.4), if and only if $v_{/\Delta}(x) = 0$ (cf. lemma 4.1 applied to the quotient valuation $v_{/\Delta}$), if and only if $v(x) \in \ker(\Gamma \to \Gamma/\Delta) = \Delta$. Thus $\Delta = v(B_\Delta^\times)$.

Thus $\Delta \mapsto B_\Delta$ is injective and order preserving. Since $\mathrm{Cnv}(\Gamma)$ is totally ordered by inclusion (cf. ordered-groups, lemma 5.2), we see that $\Delta \mapsto \Delta_B$ is an order embedding (cf. ordered-groups, lemma 1.5). It thus suffices to prove that $\Delta \mapsto B_\Delta$ it is surjective.

Suppose B is a subring of K containing A. Since $v : K^\times \to \Gamma$ is a group homomorphism and $B^\times$ is a subgroup of $K^\times$, we see immediately that $\Delta_B = v(B^\times)$ is a subgroup of Γ. To show that Δ_B is convex, we use the criterion of ordered-groups, lemma 4.5. Suppose we have $x \in B^\times$ and $\gamma \in \Gamma$ such that

$$v(x) \geqslant \gamma \geqslant 0.$$

Since $v : K^\times \to \Gamma$ is surjective, there exists $y \in K^\times$ such that $v(y) = \gamma$. Notice that $v(y) \geqslant 0$, so $y \in A \subseteq B$. Since $x \in B^\times$, we also have $y/x \in B$. But $v(x/y) = v(x) - v(y) \geqslant 0$, so $x/y \in A \subseteq B$, so in fact we have $y/x \in B^\times$. Thus $y = (y/x)x \in B^\times$, showing that $\gamma = v(y) \in \Delta_B$.

To conclude, we must show that B coincides with $B_{\Delta_B} = \{x \in K : v_{/\Delta_B}(x) \geqslant 0\}$. They are both valuation subrings of K by lemmas 1.9 and 4.1, so it is sufficient to show that one dominates the other (cf. theorem 1.4). We will show that B dominates B_{Δ_B}.

Suppose $x \in B_{\Delta_B}$. In other words, $x \in K$ and $v_{/\Delta_B}(x) \geqslant 0$. We know that $x = 0 \in B$, so suppose $x \in K^\times$. Since $v_{/\Delta_B}(x) \geqslant 0$, we see that either $v(x) \geqslant 0$ or $v(x) \in \Delta_B = v(B^\times)$ (cf. ordered-groups, lemma 6.2). In the former case, we have $x \in A \subseteq B$, and in the latter case, we have $x \in B^\times \subseteq B$ by lemma 4.4. Thus $B_{\Delta_B} \subseteq B$. Moreover, note that $x \in B^\times$ if and only if $v(x) \in \Delta_B = v(B^\times)$ (cf. lemma 4.4), if and only if $v_{/\Delta_B}(x) = 0$, if and only if $x \in B_\Delta^\times$ (cf. lemma 4.1). Thus B dominates B_{Δ_B}. □

Lemma 4.6. *Let K be a field and $v : K \to \Gamma \cup \{\infty\}$ a surjective valuation on K. Let $A = K^{v \geqslant 0}$. For a convex subgroup $\Delta \subseteq \Gamma$, we have the quotient valuation $v_{/\Delta}$ (cf. definition 3.1) and*

$$\mathfrak{p}_\Delta = \{x \in K : v_{/\Delta}(x) \gneqq 0\}$$

is a prime ideal of A. Moreover, the map $\Delta \mapsto \mathfrak{p}_\Delta$ defines an inclusion reversing bijection from $\mathrm{Cnv}(\Gamma)$ onto $\mathrm{Spec}(A)$. The inclusion reversing inverse is given by mapping a prime ideal $\mathfrak{p}$ to the convex subgroup $\Delta_\mathfrak{p} = v(A_\mathfrak{p}^\times)$.

Proof. Combine lemmas 1.9 and 4.5. □

5 Equivalence of valuations

Definition 5.1. Let A be a ring and suppose $v : A \to \Gamma \cup \{\infty\}$ and $v' : A \to \Gamma' \cup \{\infty\}$ are two valuations on A. Then v and v' are *equivalent* if, for all $a, b \in A$, we have $v(a) \geqslant v(b)$ if and only if $v'(a) \geqslant v'(b)$.

Remark 5.2. The collection of all valuations on a ring A forms a proper class, since the cardinality of the totally ordered abelian group in which a valuation takes values can be arbitrarily large. Evidently definition 5.1 defines an equivalence relation on this class. In any case, we will see momentarily that the collection of equivalence classes is a set.

Lemma 5.3. *Let A be a ring and suppose $v : A \to \Gamma \cup \{\infty\}$ and $v' : A \to \Gamma' \cup \{\infty\}$ are two valuations on A. Then the following are equivalent.*

(a) *v and v' are equivalent.*

(b) *There exists an order preserving isomorphism $f : \Gamma_v \to \Gamma_{v'}$ such that $f(v(a)) = v'(a)$ for all $a \in A$.*

(c) *We have $\operatorname{supp}(v) = \operatorname{supp}(v')$ and $\kappa(v)^{v \geqslant 0} = \kappa(v')^{v' \geqslant 0}$.*

6 Valuation spectrum

Definition 6.1. Let A be a ring. The *valuation spectrum* of A, denoted $\operatorname{Spv}(A)$, is the set of pairs $(\mathfrak{p}, B)$ where $\mathfrak{p}$ is a prime ideal in A and B is a valuation subring of $\operatorname{Frac}(A/\mathfrak{p})$. If $x = (\mathfrak{p}, B)$ is an element of $\operatorname{Spv}(A)$, we write $\operatorname{supp}(x)$ for the prime ideal $\mathfrak{p}$, $\kappa(x)$ for the field $\operatorname{Frac}(A/\mathfrak{p}(x))$, and $\kappa(x)^+$ for the valuation subring B.

Lemma 6.2. *Let A be a ring and v a valuation on A. Then v defines a unique point $x_v \in \operatorname{Spv}(A)$ such that $\operatorname{supp}(v) = \operatorname{supp}(x_v)$ and $\kappa(x_v) = \operatorname{Frac}(A/\operatorname{supp}(v))^+$.*

Lemma 6.3. *Let A be a ring and $x \in \operatorname{Spv}(A)$. There exists a valuation v on A such that $\operatorname{supp}(x) = \operatorname{supp}(v)$ and $\kappa^+(x) = \operatorname{Frac}(A/\operatorname{supp}(v))^+$.*

Proof. Let $\Gamma = \kappa(x)^\times/\kappa^+(x)^\times$ with group law written additively. Then Γ is naturally a totally ordered abelian group [Stacks, 00ID]. Consider the function $\bar{v} : \kappa(x) \to \Gamma \cup \{\infty\}$ where $\bar{v}(0) = \infty$ and $\bar{v}(x)$ is the class of x in Γ for all $x \in \kappa(x)^\times$. Then $\bar{v}$ is a valuation on $\kappa(x)$ such that

$$\kappa^+(x) = \kappa(x)^+ := \{x \in \kappa(x) : \bar{v}(x) \geqslant 0\}.$$

We denote by ν the composite $A \to \kappa(x) \to \Gamma \cup \{\infty\}$. This is a valuation on A by lemma 2.9. Clearly $\operatorname{supp}(\nu) = \operatorname{supp}(x)$ and $\kappa(x) = \operatorname{Frac}(A/\operatorname{supp}(\nu))$, and the unique extension of ν to $\kappa(x)$ as in lemma 2.10 must be $\bar{\nu}$, so

$$\operatorname{Frac}(A/\operatorname{supp}(\nu))^+ = \kappa(x)^+ = \kappa^+(x). \qquad \square$$